

MURREE, PakistanWMO: **415730**Lat: **33.917N**Lon: **73.383E**Elev: **6978**StdP: **11.35**Time Zone: **5.00 (E05)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.5	30.3	9.2	11.4	46.7	12.9	13.7	46.9	14.5	40.2	11.8	41.8	2.5	0	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	13.7	82.4	62.0	79.7	62.6	77.4	62.8	70.7	74.3	69.7	73.3	68.7	72.4	2.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	142.2	73.0	68.7	137.4	72.2	67.6	132.1	71.2	39.9	74.0	38.9	73.2	37.9	72.8	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.7	9.0	7.2	25.4	86.9	3.5	3.4	22.8	89.4	20.7	91.3	18.8	93.2	16.2	95.7
			22.5	73.2	3.4	0.7	20.1	73.7	18.1	74.1	16.2	74.4	13.7	74.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.5	41.2	42.2	50.5	58.6	65.9	69.5	68.3	67.2	65.2	60.4	53.2	46.6
	DBStd	11.17	5.71	5.80	7.92	6.68	5.42	4.68	2.95	2.11	2.88	4.31	4.70	6.05
	HDD50	768	275	226	94	10	0	0	0	0	0	2	23	138
	HDD65	3218	736	639	452	211	54	11	3	6	30	150	355	571
	CDD50	3497	4	7	109	267	494	586	568	532	456	324	118	32
	CDD65	471	0	0	1	18	83	147	106	73	36	7	0	0
	CDH74	1520	0	0	2	45	386	788	236	35	25	3	0	0
	CDH80	200	0	0	0	1	47	127	25	0	0	0	0	0
Wind	WSAvg	2.4	2.6	2.8	3.0	3.0	2.9	2.5	1.5	1.4	2.0	2.2	2.3	2.4
Precipitation	PrecAvg	69.0	5.2	5.7	5.6	5.1	3.5	5.4	13.3	12.2	6.4	2.5	1.3	2.2
	PrecMax	90.7	11.6	13.6	9.6	15.3	8.2	20.4	22.3	18.1	12.4	10.0	4.1	7.6
	PrecMin	50.6	0.0	0.3	0.6	0.9	0.9	1.5	6.4	5.9	0.9	0.0	0.0	0.0
	PrecStd	11.8	3.8	2.9	2.8	3.5	1.7	4.5	3.9	3.0	3.5	2.7	1.3	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.0	60.8	71.8	78.2	84.0	86.6	83.2	76.9	76.4	74.2	68.4	65.3
		MCWB	42.0	46.7	55.8	60.2	61.0	63.7	65.3	68.9	65.0	61.2	51.9	48.1
	2%	DB	55.3	57.3	67.6	74.7	81.0	83.4	78.9	74.9	74.4	71.9	65.3	60.7
		MCWB	41.0	43.7	53.4	57.4	59.9	62.0	67.0	68.5	65.0	59.8	50.2	45.4
	5%	DB	52.5	53.9	64.7	71.7	78.1	81.0	76.6	73.5	72.8	69.6	62.8	57.7
		MCWB	40.0	42.0	51.0	56.1	59.3	62.8	67.0	68.2	64.7	58.1	49.0	43.5
	10%	DB	49.7	51.1	61.7	68.8	75.7	78.6	74.4	72.3	71.2	67.5	60.5	54.9
		MCWB	38.4	40.7	49.0	54.1	58.1	62.8	67.1	67.9	64.4	57.2	47.9	41.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	49.3	57.4	61.8	64.8	71.3	72.2	72.0	69.5	65.6	54.6	50.7
		MCDB	54.4	58.2	70.0	74.7	77.7	76.9	75.7	74.7	73.1	70.9	64.9	62.4
	2%	WB	42.7	46.0	54.9	59.3	62.9	69.5	70.8	70.5	68.0	63.6	52.5	47.1
		MCDB	52.0	54.4	66.3	71.7	76.3	75.5	74.5	73.2	71.9	68.9	62.6	57.8
	5%	WB	41.2	43.4	52.7	57.4	61.0	67.8	69.8	69.6	66.7	61.1	50.9	45.1
		MCDB	49.8	51.8	63.4	69.7	75.0	74.5	73.4	72.4	70.8	66.9	61.0	55.3
	10%	WB	39.7	41.5	50.1	55.6	59.5	66.1	68.7	68.6	65.5	59.1	49.0	43.1
		MCDB	48.2	49.1	60.3	67.1	73.5	73.5	72.4	71.4	69.9	65.8	58.6	52.7
Mean Daily Temperature Range		MDBR	12.5	11.6	12.6	13.5	14.3	13.7	9.9	9.5	11.7	13.5	13.4	13.2
	5% DB	MCDBR	15.6	15.1	15.4	16.1	16.3	16.1	14.1	12.3	13.9	15.0	15.5	16.0
		MCWBR	10.2	10.6	10.9	11.9	11.3	11.5	10.3	10.6	12.0	12.1	11.8	11.0
		MCDBR	14.3	13.8	15.1	15.7	15.7	12.2	10.4	9.9	12.1	12.9	14.5	14.5
	5% WB	MCWBR	10.4	10.6	11.3	11.8	11.9	10.9	9.4	9.3	11.2	11.3	11.5	11.2
Clear-Sky Solar Irradiance	taub	0.319	0.357	0.366	0.380	0.396	0.448	0.495	0.474	0.403	0.367	0.342	0.316	
	taud	2.292	2.211	2.237	2.242	2.209	2.071	1.984	2.053	2.223	2.235	2.241	2.313	
	Ebn at Noon	280	279	287	287	283	267	253	257	272	272	268	273	
	Edh at Noon	33	40	42	44	46	52	57	52	42	38	34	30	
All-Sky Solar Radiation	RadAvg	898	1060	1466	1831	2178	2207	1909	1736	1698	1479	1087	880	
	RadStd	108	164	158	123	128	116	152	136	109	86	75	74	